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AI-Powered Plan Coordination

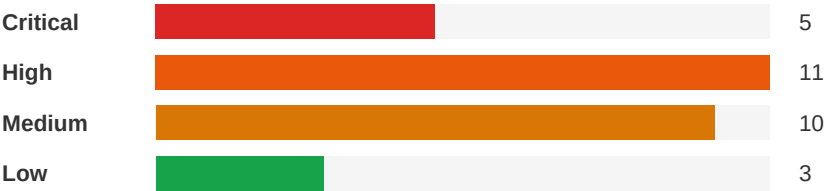
Plan Conflict & Coordination Report

Project Name:	The Atrium
Report Date:	May 14, 2026
Analyst:	Flikt.AI Automated Analysis

Executive Summary



Severity Distribution



Top Priority Actions

1. Emergency Lighting and Exit Signage Noted as 'Remain as Existing' Despite Extensive Ceiling Demolition

Severity: Critical | Electrical-Lighting ↔ Architectural

Prepare an emergency lighting and exit signage plan for all renovated corridors showing: existing emergency fixture locations to be maintained/relocated, temporary emergency lighting during construction per OSHA, reinstallation locations on new ceilings, and battery backup verification. Submit to AHJ for approval. Consider upgrading to LED emergency fixtures during renovation.

2. Missing Fire Alarm System Drawings

Severity: Critical | Architectural ↔ Electrical-Lighting

Clarify whether fire alarm design is delegated to a fire alarm contractor under separate permit. If so, document this delegation clearly on the electrical drawings. If not, provide fire alarm plans showing notification appliances, initiating devices, and FACP connections for all renovated common areas per NFPA 72 and FBC requirements for high-rise R2 occupancy.

3. No Fire Alarm Plan Provided for Renovated Common Areas — Smoke Detectors, Pull Stations, Horn/Strobes Not Shown

Severity: Critical | Electrical-Lighting ↔ Architectural

Engage a fire alarm designer to prepare a fire alarm plan for all renovated common areas. Show all initiating devices (smoke detectors, heat detectors, duct smoke detectors, pull stations, waterflow switches, tamper switches) and notification appliances (horn/strobes, speakers) per NFPA 72. Coordinate with the mechanical plans for duct detector locations and with the architectural plans for device placement relative to new walls and ceilings.

Detailed Findings

Critical

ID: C003

Emergency Lighting and Exit Signage Noted as 'Remain as Existing' Despite Extensive Ceiling Demolition

Disciplines:

Electrical-Lighting ↔ Architectural

Type:

Code Violation

Location:

All Renovated Corridors and Common Areas

Sheets:

E3, E4, A2, E2, A1

Description: Electrical keynote #1 on sheets E3 and E4 states: 'ALL EXISTING RECEPTACLES TO REMAIN AS EXISTING (REPLACED IN SAME LOCATION AS NECESSARY). NEW RECESSED LED LIGHTING FIXTURES TO BE CONNECTED TO EXISTING LIGHTING CIRCUIT SERVING AREA. TOTAL WATTAGE TO NOT INCREASE. EXISTING LIGHTING CONTROL DEVICES TO REMAIN. EMERGENCY LIGHTING/SIGNAGE LAYOUT TO REMAIN AS EXISTING.' However, the architectural RCP plans A2 through A2 call for demolition and replacement of existing ceilings with 'NEW 5/8" SUSPENDED GYPSUM BOARD CEILING' throughout corridors on all floors (4th-26th typical, PG intermediate, PH1, PH2). Demolishing existing ceilings will necessarily disturb or destroy existing emergency lighting fixtures and exit signs mounted in or on those ceilings. No plan shows the reinstallation locations of emergency fixtures after ceiling replacement. Per IBC 1008 and NFPA 101, emergency lighting must be maintained during construction and properly reinstalled. The lack of an emergency lighting plan for 22+ floors of corridor renovation is a significant life-safety gap.

Construct.

Cost

Safety

Schedule

Downstrm.

5/10

6/10

9/10

5/10

8/10

Cost Exposure: \$20,471 – \$42,285

Schedule Impact: 13 days

Recommended Action: Prepare an emergency lighting and exit signage plan for all renovated corridors showing: existing emergency fixture locations to be maintained/relocated, temporary emergency lighting during construction per OSHA, reinstallation locations on new ceilings, and battery backup verification. Submit to AHJ for approval. Consider upgrading to LED emergency fixtures during renovation.

Critical

ID: C001

Missing Fire Alarm System Drawings

Disciplines:

Architectural ↔ Electrical-Lighting

Type:

Missing Coordination

Location:

All Levels - Building-wide

Sheets:

A1, E1, E4, A2, E2, E3, COVER, E5

Description: The architectural cover sheet (A1) lists the scope of work as a complete common space interior remodel including main lobby, elevator lobbies, business center, card room, wellness center, and main corridors on each floor. The electrical notes on E1 reference smoke detectors (note #25, #29) for dwelling units but no fire alarm system drawings are provided for the common areas. For a Type I-A, R2 occupancy high-rise condominium (26+ stories per corridor plans), NFPA 72 and Florida Building Code require a building fire alarm system with pull stations, horns/strobes, smoke detectors in corridors, and waterflow/tamper monitoring. The electrical plans show only smoke detectors within the scope areas but no fire alarm notification appliances (horns/strobes), pull stations, or fire alarm control panel connections for the renovated common areas. The architectural RCP notes (A2) reference 'FOR FIRE SPRINKLER LOCATIONS, SEE FIRE PROTECTION DRAWINGS' but no fire protection drawings are included.

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CONFIDENTIAL

Construct.	Cost	Safety	Schedule	Downstrm.
				
4/10	6/10	9/10	6/10	8/10
Cost Exposure: \$17,370 – \$36,014			Schedule Impact: 10-20 days	

Recommended Action: Clarify whether fire alarm design is delegated to a fire alarm contractor under separate permit. If so, document this delegation clearly on the electrical drawings. If not, provide fire alarm plans showing notification appliances, initiating devices, and FACP connections for all renovated common areas per NFPA 72 and FBC requirements for high-rise R2 occupancy.

Critical	ID: C004			
	No Fire Alarm Plan Provided for Renovated Common Areas — Smoke Detectors, Pull Stations, Horn/Strobes Not Shown			
Disciplines:	Electrical-Lighting ↔ Architectural		Type:	Documentation Gap
Location:	All renovated common areas — Lobby, Mezzanine, Corridors (Floors 4-26)		Sheets:	E1, E2, E3, A1

Description: The project involves a complete common space interior remodel of a Type I-A, R2 occupancy high-rise condominium (26+ stories). The electrical notes on E1 reference smoke detectors (Note 25, 29) and interconnection requirements, but these notes appear to apply to dwelling units, not common areas. No fire alarm plan is provided showing: duct smoke detectors for the HVAC systems, waterflow switches for the fire sprinkler system, tamper switches, pull stations at exits, horn/strobes in corridors and common areas, or fire alarm control panel connections. Per NFPA 72 and FBC Fire Prevention Code, any renovation that alters the fire alarm notification zone layout requires updated fire alarm design. The architectural plans show altered egress paths and new wall configurations that would require repositioned fire alarm devices. This is a critical life-safety documentation gap that will prevent permit approval.

Construct.	Cost	Safety	Schedule	Downstrm.
				
7/10	8/10	10/10	8/10	9/10
Cost Exposure: \$14,400 – \$28,849			Schedule Impact: 14 days	

Recommended Action: Engage a fire alarm designer to prepare a fire alarm plan for all renovated common areas. Show all initiating devices (smoke detectors, heat detectors, duct smoke detectors, pull stations, waterflow switches, tamper switches) and notification appliances (horn/strobes, speakers) per NFPA 72. Coordinate with the mechanical plans for duct detector locations and with the architectural plans for device placement relative to new walls and ceilings.

Critical	ID: C002			
	Electrical Notes Reference NEC 2011 but Cover Sheet Cites NEC 2020 and FBC 2023			
Disciplines:	Electrical-Lighting ↔ Architectural		Type:	Code Violation
Location:	Project-wide — Electrical Notes vs. Cover Sheet Code References		Sheets:	E1, COVER

Description: The electrical notes on sheet E1 repeatedly reference NEC 2011 code sections: Note 26 cites 'NEC 2011 210-12(B)' for AFCI protection, Note 27 cites 'NEC 2011' for tamper-resistant receptacles, Note 28 references '2011 NEC 210.52(B)(C)' for countertop receptacle spacing, and Note 21 references 'NEC 250-140.' However, the architectural cover sheet (A1) explicitly states the project shall comply with 'NATIONAL ELECTRICAL CODE 2020,

FLORIDA BUILDING CODE 2023.' The NEC 2020 has significant changes from NEC 2011 including expanded AFCI requirements (210.12), expanded GFCI requirements (210.8), and updated grounding requirements. Using outdated code references could result in non-compliant installations and failed inspections by the AHJ.

Construct.	Cost	Safety	Schedule	Downstrm.
				
3/10	4/10	7/10	6/10	8/10

Cost Exposure: \$11,134 – \$24,414

Schedule Impact: 5 days

Recommended Action: Update all electrical notes on sheet E1 to reference NEC 2020 (NFPA 70-2020) and FBC 2023 code sections. Review all AFCI, GFCI, tamper-resistant, and grounding requirements against NEC 2020 to ensure compliance. Particular attention to expanded GFCI requirements in NEC 2020 Section 210.8.

Critical

ID: DISC067

Missing Fire Protection Discipline — No Fire Protection Drawings in Plan Set

Disciplines: Fire Protection ↔ General

Type: Missing Discipline

Location: Entire plan set

Sheets: 251210, A2, A1, A-101_1, A-101_2, A-101_3

Description: The plan set contains 5/9 expected disciplines for a retail project. Fire Protection drawings are entirely absent. For retail projects, Fire Protection is a required discipline. This may indicate missing drawings, a separate submittal package, or a scope exclusion that should be documented.

Construct.	Cost	Safety	Schedule	Downstrm.
				
5/10	5/10	8/10	5/10	4/10

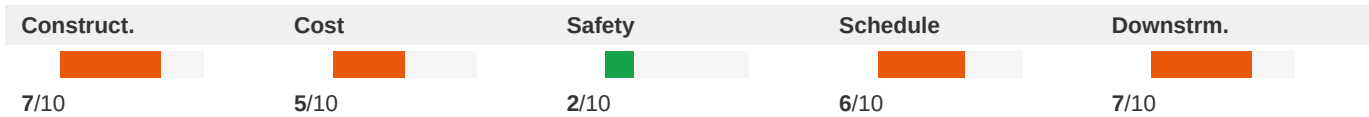
Cost Exposure: \$8,647 – \$19,035

Schedule Impact: 9 days

Recommended Action: Verify whether Fire Protection drawings exist and were omitted from the plan set, or if a separate submittal is expected. If the scope intentionally excludes Fire Protection, document this exclusion in the project specifications or general notes.

ID: C014

High

Architectural RCP Ceiling Heights vs. Mechanical Duct Sizes - Insufficient Clearance Verification**Disciplines:** Architectural ↔ Mechanical**Type:** Missing Coordination**Location:** Lobby Level - Corridor, Cafe Lounge, Business Center**Sheets:** A2, M1**Description:** (See plan set for details)**Cost Exposure:** \$6,500 – \$14,535**Schedule Impact:** 7 days

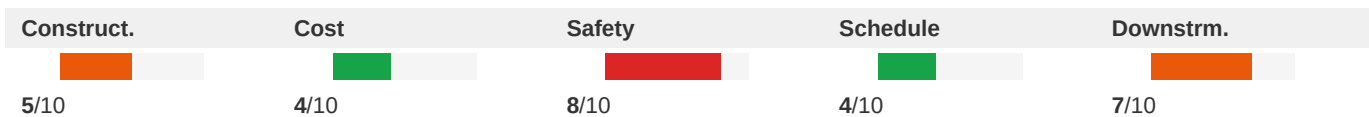
Recommended Action: Provide a composite ceiling coordination drawing or building section showing slab-to-slab height, duct routing, sprinkler piping, lighting fixtures, and finish ceiling elevation. Verify minimum 24" plenum space is available in all areas. Consider reducing duct depth in tight areas by using wider, shallower duct profiles.

ID: C009

High

Mechanical Duct Smoke Detector Requirement Not Shown on Electrical/Fire Alarm Plans**Disciplines:** Mechanical ↔ Electrical-Lighting**Type:** Missing Coordination**Location:** Lobby Level and Mezzanine - HVAC Ductwork**Sheets:** M3, M1, M2, E1

Description: Mechanical HVAC notes on M3 reference NFPA 90A and 90B compliance (Note 8: 'ALL HVAC WORK SHALL BE IN ACCORDANCE WITH NFPA 90A, 90B, 96, 54'). NFPA 90A requires duct smoke detectors on supply air systems with fan capacity over 2000 CFM. The lobby mechanical plan (M1) shows extensive new ductwork with multiple supply runs serving the lobby, cafe lounge, business center, and other spaces. The combined CFM values shown on M1 (e.g., 250 CFM, 300 CFM, 200 CFM across many diffusers) suggest the total system capacity likely exceeds 2000 CFM. However, no duct smoke detectors are shown on any electrical plan (E1-E5), and there is no fire alarm plan in the drawing set. The mechanical notes also state (Note 14): 'PROVIDE ALL FANS AND ROOFTOP UNITS WITH RELAYS TO SHUT DOWN WHEN FIRE ALARM IS INITIATED. COORDINATE LOCATION WITH THE ELECTRICAL' and (Note 15): 'IN THE EVENT OF FAN SHUT DOWN, ALL DUCT MOUNTED DETECTORS SHALL REMAIN IN OPERATION' - explicitly referencing duct-mounted detectors that do not appear on any electrical drawing.

**Cost Exposure:** \$6,297 – \$14,387**Schedule Impact:** 5-7 days

Recommended Action: Add duct smoke detectors to the electrical/fire alarm drawings at all supply duct locations where required by NFPA 90A (systems over 2000 CFM). Coordinate detector locations with mechanical ductwork layout. Ensure wiring to fire alarm control panel is shown. Verify fan shutdown relay connections are detailed on electrical plans.

High

ID: C013

Corridor Ceiling Heights on RCP vs. Available Plenum Space for New Ductwork — 4th-26th Floor Typical

Disciplines:	Architectural ↔ Mechanical	Type:	Specification Mismatch
Location:	4th-26th Floor Typical Corridors	Sheets:	A2, E4

Description: The architectural RCP for the 4th-26th typical corridor (A2) shows a finish ceiling height of 7'-6" with new 5/8" suspended gypsum board ceiling. The electrical corridor plans (E4) show new recessed LED lighting fixtures to be connected to existing circuits. The mechanical plans show existing A/C systems to remain for corridor areas. With a typical concrete slab-to-slab height in a high-rise of approximately 9'-0" to 9'-6", a 7'-6" finish ceiling leaves only 18" to 24" of plenum space for existing ductwork, sprinkler piping, electrical conduit, and the new suspended ceiling framing. The architectural notes state 'FINISH CEILING HEIGHTS ARE GIVEN FROM TOP OF CONCRETE' — if this means from top of the floor slab below, the actual plenum space above the ceiling may be insufficient for all MEP systems, especially where new ductwork transitions occur.

Construct.	Cost	Safety	Schedule	Downstrm.
7/10	4/10	3/10	5/10	6/10

Cost Exposure: \$5,562 – \$12,880	Schedule Impact: 5 days
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Recommended Action: Verify the slab-to-slab dimension in the existing building. Confirm that the 7'-6" ceiling height provides adequate plenum space (minimum 12" typically required) for existing ductwork, sprinkler heads, and new lighting fixtures. Prepare a ceiling coordination section drawing showing all MEP elements within the plenum space.

High

ID: C005

Drawing Index: Sheet A6 missing from bundle

Disciplines:	General ↔ General	Type:	Missing Coordination
Location:	COVER (cover, p.1)	Sheets:	A6

Description: 12 sheets appear in the uploaded bundle but are not listed on the cover-sheet drawing index: E1, E1, E2, E3, E4, E5, P3, M3, P1, P2 and 2 more. Index appears stale with respect to the issued set.

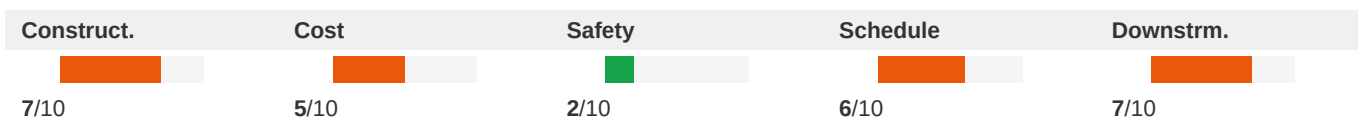
Construct.	Cost	Safety	Schedule	Downstrm.
5/10	5/10	6/10	5/10	3/10

Cost Exposure: \$5,263 – \$12,277	Schedule Impact: 13 days
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Recommended Action: Confirm with design team whether A6 is to be issued or removed from the drawing index.

ID: C012

High

Architectural RCP Ceiling Heights vs Mechanical Duct Sizes - Lobby Plenum Congestion**Disciplines:** Architectural ↔ Mechanical**Type:** Spatial Clash**Location:** Lobby Level - Main Lobby, Corridors, Business Center**Sheets:** A2, M1**Description:** (See plan set for details)**Cost Exposure:** \$5,074 – \$11,873**Schedule Impact:** 5-10 days

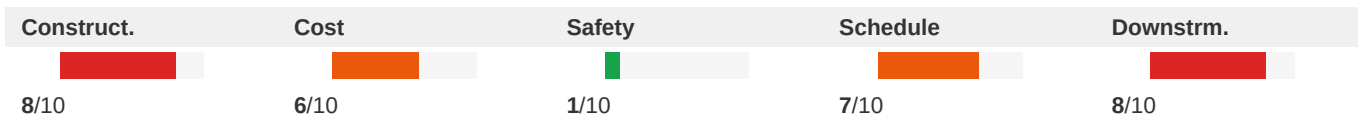
Recommended Action: Verify floor-to-floor height in the existing building. Calculate available plenum space in each area (floor-to-floor minus finished ceiling height). Confirm all ductwork can fit within the available plenum with minimum 2" clearance above the ceiling grid. Pay special attention to corridor areas where multiple duct runs converge. Consider reducing duct height in tight areas by increasing width.

ID: C011

High

Interior Design Package Referenced but Not Included — Finish Specifications Undefined**Disciplines:** Architectural ↔ Architectural**Type:** Documentation Gap**Location:** Project-wide — All renovated spaces**Sheets:** A1, A2

Description: Multiple architectural sheets reference an 'Interior Design Package' that is not included in the construction document set. Proposed plan keynote #2 on A1 states 'REPLACE EXISTING FLOORING W. NEW AND G.C. INSTALL WHISPERMAT SOUND CONTROL MEMBRANE WITH AN STC OF 72 UNDER THE FLOORING (REFER INTERIOR DESIGN PACKAGE FOR FINISHES).' Keynote #3 states 'NEW WALL FINISHES (REFER INTERIOR DESIGN PACKAGE FOR FINISHES).' RCP sheets A2 and A2 state 'REPLACE EXISTING CEILING W. NEW (REFER INTERIOR DESIGN PACKAGE FOR FINISHES).' Without this package, contractors cannot price or procure floor finishes, wall finishes, ceiling finishes, or specialty items. This affects all trades for coordination of substrate preparation, adhesive compatibility, and installation methods.

**Cost Exposure:** \$4,923 – \$10,853**Schedule Impact:** 14-21 days

Recommended Action: Issue the Interior Design Package as a formal addendum to the construction documents before bidding. Include finish schedules, material specifications, tile layouts, paint colors, wallcovering specifications, and any specialty finish details. All trades need this information for accurate pricing and coordination.

ID: C008

High

No Mechanical HVAC Design for Mezzanine Gym, Massage, and Wellness Spaces**Disciplines:** Architectural ↔ Mechanical**Type:** Missing Coordination**Location:** Mezzanine Level - Gym, Massage Rooms, Wellness Areas**Sheets:** A1, M2, E3, P2, A-101_1, P1

Description: Architectural plan A1 shows extensive renovation of the mezzanine level including a new Gym, Massage rooms, Men's/Women's Restroom/Lockers, Steam, Sauna, Corridor, and Unit spaces. The mechanical mezzanine plan M2 shows only limited ductwork serving a small portion of the mezzanine with an existing A/C1 system to remain. The gym area, which is a high-occupancy space requiring significant ventilation and cooling (per ASHRAE 62.1, gyms require 0.06 CFM/sf plus 20 CFM/person), has no new HVAC ductwork shown. The massage rooms, locker rooms, and wellness spaces also lack dedicated supply and return air. The mechanical plan only notes 'EXISTING A/C1 SYSTEM TO REMAIN AND REUSE. CONTRACTOR TO VERIFY EXACT LOCATION, CAPACITY, AND CONDITION.' There is no verification that the existing system capacity is adequate for the renovated spaces with potentially different occupancy loads.

Construct.	Cost	Safety	Schedule	Downstrm.
				
6/10	6/10	4/10	5/10	7/10

Cost Exposure: \$4,920 – \$10,745**Schedule Impact:** 14 days

Recommended Action: Perform a load calculation for the renovated mezzanine spaces per ASHRAE standards. Verify existing A/C1 system capacity against new loads. Design supplemental HVAC if existing system is inadequate. Add supply/return ductwork to gym, massage, locker rooms, and wellness areas. Provide exhaust for steam room, sauna, and locker rooms per FBC Mechanical.

ID: C006

High

Sauna and Steam Room Electrical Power Not Shown on Panel Schedule**Disciplines:** Architectural ↔ Electrical-Lighting**Type:** Missing Coordination**Location:** Mezzanine Level - Sauna and Steam Rooms**Sheets:** A1, E3, E5, A-101_1

Description: Architectural plans A1 (Proposed Mezzanine Level) and A1 (Demo Mezzanine) show sauna rooms and steam rooms as part of the wellness/spa area renovation. The demolition plan notes 'DEMO EXISTING STEAM REPLACE W NEW' and 'REPLACE EXISTING STEAM' indicating new steam generators are being installed. The electrical mezzanine plan E3 shows the area but the panel schedule on E5 (Panel-LAA) only lists 'EXISTING STEAM GEN. (NEW)' and 'EXISTING SAUNA (NEW)' circuits at 2,200 VA each. Steam generators and saunas for commercial spa applications typically require 9-18kW (significantly more than 2,200 VA). The 2,200 VA rating appears to be the existing circuit capacity being reused, which may be inadequate for new commercial-grade equipment. The panel-LAA demand load calculation shows only 32,350 VA connected load on a 100A panel — if the steam and sauna loads are actually 9-18kW each, the panel would be significantly overloaded.

Construct.	Cost	Safety	Schedule	Downstrm.
				
5/10	5/10	6/10	4/10	6/10

Cost Exposure: \$4,528 – \$10,448**Schedule Impact:** 7-10 days

Recommended Action: Verify actual kW ratings of new steam generator and sauna heater equipment. Compare against Panel-LAA circuit ratings (currently shown as 2,200 VA each). If new equipment exceeds existing circuit capacity, upsize breakers, conductors, and potentially the panel bus. Coordinate with mechanical/plumbing for steam generator specifications.

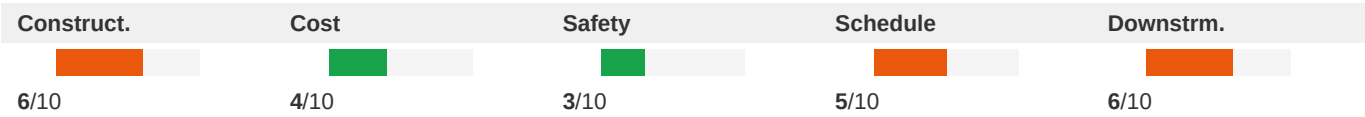
High

ID: C010

No Condensate Drain Lines Shown for Existing A/C Systems Serving Renovated Spaces

Disciplines:	Mechanical ↔ Plumbing	Type:	Missing Coordination
Location:	Lobby and Mezzanine Levels — Multiple A/C systems	Sheets:	M1, M2, P1, P2

Description: The mechanical plans (M1, M2) show multiple existing A/C systems to remain and reuse (A/C1, A/C1, A/C1, A/C2) with new ductwork connections. The plumbing notes on the Arch/Plumbing sheet state 'THE PLUMBING CONTRACTOR SHALL RUN BUILDING CONDENSATE RISER WITH STUB-OUTS AT THE AIR CONDITIONING UNITS IF APPLICABLE. THE MECHANICAL CONTRACTOR IS RESPONSIBLE TO CONNECT HIS EQUIPMENT TO THOSE.' However, neither the mechanical plans nor the plumbing plans show condensate drain routing for these existing systems. With new ceiling and floor finishes being installed throughout (per architectural keynotes), the condensate drain paths need to be verified and potentially rerouted to accommodate the renovation. No condensate receptors or drain connections are shown on the plumbing plans for the lobby or mezzanine mechanical equipment.



Cost Exposure: \$4,227 – \$10,202

Schedule Impact: 3-5 days

Recommended Action: Mechanical engineer should show condensate drain routing on the mechanical plans for all A/C units. Plumbing engineer should coordinate condensate receptor locations and tie-in points to the sanitary system. Verify existing condensate paths are not disrupted by the renovation work.

High

ID: C007

Panel-LA Demand Load Exceeds Main Breaker Capacity

Disciplines:	Electrical-Lighting ↔ Electrical-Lighting	Type:	Specification Mismatch
Location:	Electrical Room - Lobby Level	Sheets:	E1, E5

Description: Panel-XB is specified as 100A bus, 120Y/208V, 3-phase, 4-wire with AIC=10K. The demand load calculations show: RECPT 1st 10,000 VA @ 100% = 3,000 VA, REST @ 50% = 3,000 VA, LIGHTING LOAD @ 125% = 10,000 VA (but listed as 12,500 VA at 125%), LARGEST MOTOR @ 125% = 1,200 VA, OTHER MOTORS @ 100% = 1,500 VA, resulting in TOTAL LOAD = 21,000 VA. At 208V 3-phase, 21,000 VA / (208 × 1.732) = 58.3A, which fits within 100A bus. However, the connected load subtotals show 18,750 VA on one side and 5,900 VA on the other, totaling approximately 26,250 VA connected. The panel schedule shows multiple new circuits being added (supply fan, exhaust fans, front door operators, fire smoke dampers) alongside existing loads. The notes state 'VERIFY ALL EQUIPMENT LOAD AND BREAKER AND WIRE SIZES PRIOR TO INSTALLATIONS' — indicating the engineer is uncertain about final loads. With 78A calculated demand on a 100A panel fed through an existing system, there is minimal spare capacity for any additional loads discovered during construction.

Construct.	Cost	Safety	Schedule	Downstrm.
				
4/10	5/10	5/10	4/10	6/10

Cost Exposure: \$4,164 – \$9,445

Schedule Impact: 5-10 days

Recommended Action: Perform a detailed load calculation with actual equipment nameplate data before finalizing panel schedule. Consider whether Panel-LA has adequate spare capacity for all new loads. If loads exceed 80% of panel rating (320A continuous), evaluate upsizing the panel or redistributing loads. Verify feeder sizing from main switchboard to Panel-LA is adequate.

High

ID: DRN001

Roof Drain Without Downspout: ROOF DRAIN has no routing

Disciplines: A ↔ P

Type: Drainage Gap

Location: See sheet A2

Sheets: A2, L'Her

Description: Architectural plan shows roof drain 'AGE' on sheet BHE - 250305 - L'Her. Plumbing plans contain no downspout or storm leader routing from this drain location.

Construct.	Cost	Safety	Schedule	Downstrm.
				
5/10	5/10	3/10	5/10	6/10

Cost Exposure: \$3,751 – \$8,596

Schedule Impact: 1

Recommended Action: Coordinate with plumbing: add storm leader from ROOF DRAIN to appropriate grade discharge.

Medium

ID: C017

ADA 604.5.1: Grab Bar Height (34 instances)

Disciplines: Electrical-Lighting ↔ Architectural

Type: Code Violation

Location: Sheet A-101_1_p1

Sheets: A-101_1, A2, A7

Description: Element 'grab_bar' (ID: grab_bar_A-101_1_p1_1) at Sheet A-101_1_p1: height=79.0" exceeds ADA maximum 36.0". ADA 2010 §604.5.1 (Grab Bar Height) requires reach range 33.0"–36.0" or maximum 36.0". Measured value does not comply. AGGREGATED FINDING: 34 individual instances detected across sheets: A-101_1_p1, A2, A7, A7. Equipment/element tags: .

Construct.	Cost	Safety	Schedule	Downstrm.
				
5/10	5/10	3/10	6/10	3/10

Cost Exposure: \$2,126 – \$5,215

Schedule Impact: 12 days

Recommended Action: Revise grab_bar installation to meet ADA §604.5.1 minimum 33.0" requirement. Coordinate with accessibility consultant.

Medium

ID: CV2025

No fire protection sheets — NFPA 72 fire alarm requirements may not be addressed

Disciplines:	Fire Protection ↔ Electrical	Type:	Code Violation
Location:	Project-level discipline coverage	Sheets:	251210, 2025.12.09_Mech, 2025.10.28_Plum

Description: Project includes MEP disciplines but no dedicated fire protection sheets. NFPA 72 requires fire alarm and notification appliance documentation. Verify fire protection design is included in the plan set or is being provided under a separate permit.

Construct.	Cost	Safety	Schedule	Downstrm.
3/10	4/10	6/10	3/10	2/10

Cost Exposure: \$2,021 – \$5,151	Schedule Impact: 3-7 days
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Recommended Action: Confirm fire protection sheets are included in the complete plan set, or verify fire protection is under a separate permit with its own plan review.

Medium

ID: C021

Florida Amendment Violation: Electrical cites NEC only, missing FBC Electrical amendments

Disciplines:	E ↔ E	Type:	Ahj Amendment
Location:	See sheets E1	Sheets:	E1

Description: Project in FL. Plans reference 'NEC 2011' without citing Florida amendments to NEC (FBC Electrical / Ch. 27). Florida requires AFCI protection in all habitable rooms (broader than NEC base) and enhanced tamper-resistant receptacle requirements.

Construct.	Cost	Safety	Schedule	Downstrm.
4/10	4/10	2/10	3/10	6/10

Cost Exposure: \$1,930 – \$4,762	Schedule Impact: 3
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Recommended Action: Verify electrical compliance against FBC Electrical (Ch. 27) amendments to NEC. Florida extends AFCI requirements to all habitable rooms and has additional tamper-resistant receptacle requirements beyond base NEC.

ID: DRN003

Medium

Overflow Drain Missing: 12 primary drain(s), no secondary/overflow drains shown**Disciplines:** A ↔ A**Type:** Drainage Gap**Location:** See sheets A2, BHE - 250305 - L'Her, A6**Sheets:** A2, L'Her, A6, A7

Description: Architectural plans show 12 primary roof drain(s) but no secondary overflow drains or scuppers. IPC 1504.4 requires secondary (emergency) roof drainage. Verify that overflow/scupper drainage is provided for all roof drain areas.

Construct.	Cost	Safety	Schedule	Downstrm.
				
3/10	4/10	3/10	2/10	5/10

Cost Exposure: \$1,812 – \$4,660**Schedule Impact:** 1

Recommended Action: Verify overflow drainage per IPC 1504.4: add secondary roof drains or scuppers with independent discharge path. Coordinate with architect and plumbing.

ID: C019

Medium

Exhaust Fan Electrical Circuits Not Coordinated with Mechanical Exhaust Layout**Disciplines:** Mechanical ↔ Electrical-Lighting**Type:** Missing Coordination**Location:** Mezzanine Level - Restrooms, Locker Rooms**Sheets:** M2, E3, E5, M1, E1

Description: The mezzanine mechanical plan (M2) shows exhaust ductwork serving the restrooms and locker room areas with specific CFM requirements. The electrical mezzanine plan (E3) shows bathroom exhaust fan symbols in some locations, but the panel schedule (E5, Panel-LAA) only lists 'EXHAUST FAN MEZZ. (EXISTING)' as a single existing circuit. The mechanical plan shows new ductwork connections that may require new or upgraded exhaust fans. The general mechanical notes state exhaust outlets shall be farther than 3 feet from operable windows and exhaust ductwork requires specific slope installation. There is no clear coordination between the new mechanical exhaust design and the electrical circuits powering the exhaust fans — the electrical drawings appear to rely on existing circuits while mechanical shows new ductwork configurations.

Construct.	Cost	Safety	Schedule	Downstrm.
				
4/10	3/10	2/10	4/10	5/10

Cost Exposure: \$1,777 – \$4,639**Schedule Impact:** 3-4 days

Recommended Action: Coordinate exhaust fan sizing between mechanical and electrical disciplines. Verify that existing exhaust fan circuits on Panel-LAA are adequate for any new or relocated exhaust fans shown on the mechanical plans. Add new circuits to the panel schedule if additional exhaust fans are required. Show exhaust fan electrical connections on the electrical mezzanine plan (E3).

Medium

ID: C016

Architectural Bar Area Shown but No Plumbing Sink Connection Detailed**Disciplines:** Architectural ↔ Plumbing**Type:** Missing Coordination**Location:** Lobby Level - Bar / Cafe Lounge 3 Area**Sheets:** A1, P1, P2

Description: Architectural plan A1 shows replacement of existing steam room ('REPLACE EXISTING STEAM REPLACE W NEW') and new sauna on the mezzanine level. Steam generators require cold water supply, condensate/steam drain, and typically a floor drain. The mezzanine plumbing plan P2 shows only a cold water tie-in and sanitary tie-in for what appears to be a relocated sink area - no plumbing connections are shown for the steam generator or floor drains in the steam room or sauna areas. Steam generators typically require: 1/2" minimum cold water supply, steam line to steam head, condensate drain, and floor drain in the steam room. None of these are detailed on the plumbing plans. The plumbing scope note on P2 indicates 'NOT IN SCOPE' for large portions of the mezzanine.

Construct.	Cost	Safety	Schedule	Downstrm.
				
4/10	3/10	1/10	4/10	5/10

Cost Exposure: \$1,758 – \$4,611**Schedule Impact:** 3-5 days

Recommended Action: Verify if the new bar area requires a sink or other plumbing fixtures. If so, add plumbing connections (hot water, cold water, sanitary drain, and vent) to the plumbing plans for the bar location. Coordinate bar fixture requirements with the architectural interior design package and owner. If no bar sink is required, add a clarifying note.

Medium

ID: C015

Instant Water Heater Electrical Circuit Sizing Mismatch**Disciplines:** Plumbing ↔ Electrical-Lighting**Type:** Specification Mismatch**Location:** Lobby Level - Coffee Maker / Pantry Area**Sheets:** P1, E1, E5

Description: The plumbing fixture schedule specifies a Titan N4 instantaneous electric water heater rated at 4.2kW, 208V, 1.0 GPM recovery at 33°F temperature rise. The electrical panel schedule on E5 shows Panel-LA Section 1&2 with an 'INSTANT WATER HEATER (NEW)' circuit at 2,100 VA on a single-pole 20A breaker. However, a 4.2kW (4,200W) load at 208V draws approximately 20.2A, which is at the limit of a 20A breaker (NEC requires load not to exceed 80% of breaker rating for continuous loads = 16A). If this water heater operates continuously for more than 3 hours (likely during peak use), the 20A breaker is undersized. Additionally, the panel schedule shows this as a single-pole circuit (120V), but the Titan N4 requires 208V — this is a voltage mismatch that will prevent the unit from operating correctly.

Construct.	Cost	Safety	Schedule	Downstrm.
				
3/10	2/10	5/10	2/10	4/10

Cost Exposure: \$1,278 – \$3,510**Schedule Impact:** 2-3 days

Recommended Action: Verify the instant water heater circuit on Panel-LA is configured for 208V single-phase (2-pole breaker) with minimum 30A breaker and #10 AWG copper conductors per NEC 422.11. Coordinate between plumbing and electrical contractors to confirm voltage, amperage, and wire sizing match the Titan N4 nameplate requirements.

Medium

ID: G026

Missing Gas Supply for DISHWASHER — Mechanical scope gap**Disciplines:** Plumbing ↔ Mechanical**Type:** Missing Coordination**Location:** P1**Sheets:** P1

Description: DISHWASHER found on Plumbing plans (sheet P1) but no corresponding gas supply found on Mechanical plans. Kitchen equipment may require gas service — verify fuel source

Construct.	Cost	Safety	Schedule	Downstrm.
				
3/10	4/10	2/10	2/10	3/10

Cost Exposure: \$1,352 – \$3,369**Schedule Impact:** 3 days

Recommended Action: Verify Mechanical plans include gas supply for DISHWASHER. Issue RFI if missing.

Medium

ID: C020

Mechanical Plan Shows Spaces Not Covered by Electrical Scope**Disciplines:** Mechanical ↔ Electrical-Lighting**Type:** Missing Coordination**Location:** Lobby Level - Wellness/Pilates Room, Cafe Lounge, Orchid Room**Sheets:** M1, E1, E2, M2, M3, E3

Description: The mechanical lobby plan M1 shows new ductwork and air distribution for the Wellness Pilates Room (room 143), Cafe Lounge 3 (room 134), Orchid Room 2 (room 132), and other spaces with specific CFM values and new linear diffusers. These spaces are receiving new HVAC systems that require thermostat wiring and control connections. Electrical note 17 on E1 states 'ELECTRICAL CONTRACTOR SHALL PROVIDE CONTROL WIRES BETWEEN A.H.U. AND C.U.' and note 24 states 'RUN ALL C.U. LOW VOLTAGE WIRES FROM A.H.U. TO C.U. IN 1/2" CONDUIT.' However, the electrical plans E1 and E2 do not clearly show thermostat locations or low-voltage control wiring routes for the new HVAC zones in these spaces. The mechanical notes on M3 state (Note 13): 'ALL LOW VOLTAGE WIRING AND CONDUIT REQUIRED FOR MECHANICAL EQUIPMENT SHALL BE FURNISHED AND INSTALLED BY MECHANICAL CONTRACTOR' which partially addresses this, but coordination of thermostat locations with architectural wall layouts is still needed.

Construct.	Cost	Safety	Schedule	Downstrm.
				
4/10	2/10	1/10	3/10	5/10

Cost Exposure: \$983 – \$2,502**Schedule Impact:** 2-4 days

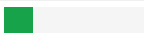
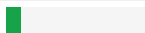
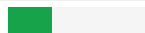
Recommended Action: Coordinate thermostat locations between mechanical and electrical disciplines. Show thermostat locations on electrical plans or confirm mechanical contractor scope includes all low-voltage wiring. Verify thermostat locations do not conflict with architectural wall finishes or casework.

Medium

ID: C018

Architectural Cove Lighting J-Box Coordination with Electrical**Disciplines:** Architectural ↔ Electrical-Lighting**Type:** Missing Coordination**Location:** Lobby Level - Lobby, Reception, Corridors**Sheets:** A5, A2, E2, E5

Description: Architectural RCP sheet A2 includes notes 'COVE LIGHT PROVIDE J-BOX' at multiple locations throughout the lobby level, and detail sheet A5 shows a 'COVE LIGHTING SOFFIT DETAIL' with LED light and neon LED light specifications. However, the electrical lighting plan (E2) shows circuit designations (XB2, LA2, XL1, etc.) for recessed fixtures and other lighting but does not clearly show dedicated circuits or junction boxes for the cove lighting elements called out on the RCP. The lighting keynote #1 on E2 states 'REPLACE EXISTING FIXTURES WITH LED EQUIVALENT. COORDINATE EXACT MODEL WITH OWNER. TOTAL WATTAGE TO REMAIN THE SAME.' This suggests the electrical scope is limited to fixture replacement, while the architectural scope includes new cove lighting that would require additional circuits and wiring not currently shown.

Construct.	Cost	Safety	Schedule	Downstrm.
				
4/10	2/10	1/10	3/10	4/10

Cost Exposure: \$973 – \$2,488

Schedule Impact: 2-3 days

Recommended Action: Coordinate cove lighting junction box locations between architectural details (A5) and electrical plan (E2). Confirm circuit assignments for all cove lighting on the panel schedule. Verify total wattage of new cove lighting does not exceed existing circuit capacity or add new circuits as needed.

Low

ID: C022

Plumbing Relocated Sink Not Reflected on Electrical Power Plan

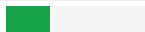
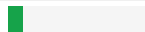
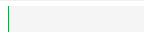
Disciplines: Plumbing ↔ Electrical-Lighting

Type: Missing Coordination

Location: Lobby Level - Coffee Maker Area

Sheets: P1, E1, E5

Description: The plumbing plan P1 notes 'REPLACE EXISTING DRINKING FOUNTAIN WITH NEW MODEL EQUIVALENT TO EXISTING IN MANNER. VERIFY EXACT LOCATION AND CONDITION.' The plumbing fixture schedule specifies an Elkay EZSTL8LC Hi-Lo electric water cooler rated at '8.0 GPH 50°F DRINKING WATER BASED ON 80°F INLET WATER AND 90°F AMBIENT.' This is a refrigerated drinking fountain requiring a dedicated 115V electrical circuit. The electrical power plan E1 does not show a receptacle or dedicated circuit at the drinking fountain location. The panel schedules on E5 do not list a circuit for a drinking fountain or water cooler. The existing unit may have had an existing circuit, but with the renovation scope, this needs to be verified and shown on the electrical plans.

Construct.	Cost	Safety	Schedule	Downstrm.
				
4/10	4/10	3/10	1/10	0/10

Cost Exposure: \$558 – \$1,393

Schedule Impact: 1-2 days

Recommended Action: Add a dedicated 208V circuit and disconnect/connection point on the electrical power plan (E1) at the relocated tankless water heater location shown on the plumbing plan (P1). Coordinate exact mounting location between plumbing and electrical contractors. Ensure the circuit is properly identified on the panel schedule.

Low

ID: C024

Door Swing Clearance Conflict: door near PANEL FLUSH**Disciplines:** E ↔ A**Type:** Clearance Obstruction**Location:** See plan set**Sheets:** P1

Description: Door swing detected in proximity to PANEL FLUSH. NEC 110.26 requires 36" clear working space. Door swing arc may encroach on required clearance zone.

Construct.	Cost	Safety	Schedule	Downstrm.
				
2/10	1/10	2/10	3/10	1/10

Cost Exposure: \$531 – \$1,373**Schedule Impact:** 1

Recommended Action: Verify door swing direction does not obstruct panel working space. Consider reversing door swing or relocating panel.

Low

ID: C023

Drinking Fountain Replacement Not Coordinated with Electrical**Disciplines:** Plumbing ↔ Electrical-Lighting**Type:** Missing Coordination**Location:** Lobby Level - Near Corridor**Sheets:** P1, E1, E5

Description: The plumbing plan P1 shows 'REPLACE EXISTING DRINKING FOUNTAIN WITH NEW' with an Elkay EZSTL8LC Hi-Lo electric water cooler specified in the fixture schedule (8.0 GPH, 50°F drinking water). This is an electrically-powered refrigerated water cooler requiring a dedicated 115V circuit. The electrical power plan E1 does not clearly show a dedicated circuit or receptacle for the replacement drinking fountain. The panel schedule on E5 does not appear to have a specific circuit identified for the drinking fountain/water cooler, though it may be served by an existing circuit. Since this is a replacement unit, the existing electrical connection may suffice, but the new unit's electrical requirements should be verified against the existing circuit.

Construct.	Cost	Safety	Schedule	Downstrm.
				
2/10	1/10	1/10	1/10	3/10

Cost Exposure: \$525 – \$1,365**Schedule Impact:** 1-2 days

Recommended Action: Verify that the existing electrical circuit serving the current drinking fountain is adequate for the new Elkay EZSTL8LC unit. Confirm circuit amperage and receptacle type match the new unit's requirements. If the new unit has different electrical requirements, add or modify the circuit on the electrical plans and panel schedule.

Methodology & Disclaimer

Analysis Methodology

This conflict analysis was performed using Flikt.AI's proprietary AI-powered plan coordination engine. The system analyzed architectural and engineering plans to identify potential spatial clashes, coordination issues, and conflicts that could impact project delivery. Each conflict has been evaluated across five key dimensions: constructability, cost impact, safety implications, schedule disruption, and downstream effects.

Confidence Levels

Each identified conflict includes a confidence score (0–1.0) indicating detection reliability. Scores of 0.9+ indicate high certainty; 0.7–0.9 indicate good confidence; below 0.7 warrants manual verification. Human review is recommended for all critical and major severity conflicts.

Cost & Schedule Estimates

Cost exposure ranges represent estimated rework, delay, and mitigation expenses. Schedule impact estimates indicate typical delay periods based on conflict severity and discipline involvement. Actual costs depend on project-specific factors, contractual arrangements, and resolution strategies.

Important Disclaimer

Flikt.AI flags discrepancies; resolution is the responsibility of the Architect/Engineer of record. This report represents an automated analysis by Flikt.AI and should not be considered a complete or exhaustive identification of all plan conflicts. The report is intended to supplement, not replace, traditional coordination meetings and detailed plan reviews by qualified professionals. Flikt.AI makes no warranty regarding completeness or accuracy. Project teams should verify all findings before making decisions based on this analysis. Use of this report is subject to Flikt.AI's terms of service.

About Flikt.AI

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Flikt.AI is the leading AI platform for coordinating complex construction projects, identifying conflicts early, and improving project delivery outcomes.

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